

ZIMNICEA, Romania

WMO: 154980

Lat: 43.6614N

Lon: 25.3536E

Elev: 114

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	16.1	8.4	8.5	11.1	13.7	11.1	16.7	27.1	33.2	21.2	37.8	3.8	250	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	94.8	71.6	92.1	70.1	89.4	69.5	74.6	88.4	73.1	86.3	71.8	84.3	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.7	113.4	79.9	69.2	107.8	78.0	68.0	103.2	76.1	38.1	88.2	36.8	86.0	35.6	83.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	15.9	13.2		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	30.0	38.3	46.2	54.2	64.2	72.1	75.1	75.2	67.3	54.4	43.6	36.0
	DBStd	16.82	9.00	8.64	7.60	7.15	5.67	4.85	4.50	4.77	6.37	7.11	8.52	6.98
	HDD50	1861	621	334	164	40	0	0	0	0	0	39	229	434
	HDD65	4757	1086	747	584	328	82	5	1	2	50	331	643	898
	CDD50	3616	0	7	45	166	439	664	778	782	520	177	36	2
	CDD65	1037	0	0	0	5	56	219	314	319	120	4	0	0
	CDH74	11221	0	0	11	154	662	2151	3295	3581	1260	102	5	0
	CDH80	4793	0	0	0	25	162	846	1495	1777	471	17	0	0
	WSAvg	4.6	4.6	4.9	5.2	5.5	4.6	4.4	4.3	4.0	4.5	4.2	3.8	5.4
	PrecAvg	22.2	1.7	1.3	1.5	1.7	2.4	2.4	2.0	1.6	2.2	1.9	1.4	1.9
Precipitation	PrecMax	35.7	4.7	3.9	3.1	3.3	5.4	6.7	6.6	5.1	6.5	4.9	3.9	6.2
	PrecMin	13.1	0.1	0.1	0.0	0.1	0.6	0.2	0.3	0.2	0.2	0.0	0.0	0.2
	PrecStd	5.4	1.0	0.8	0.8	0.8	1.1	1.5	1.6	1.3	1.6	1.3	0.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	67.9	75.5	83.4	90.2	96.4	98.4	98.2	93.4	81.6	72.2	60.5
		MCWB	47.0	57.6	59.3	63.9	69.8	74.9	76.1	71.2	67.6	62.1	59.6	52.3
	2%	DB	49.1	61.1	69.9	78.6	84.6	90.9	94.3	95.2	88.9	76.5	65.8	54.9
		MCWB	43.8	51.6	55.1	61.6	67.2	71.8	72.1	71.0	66.2	61.2	57.0	48.0
	5%	DB	44.7	56.0	65.1	73.3	81.7	88.1	91.4	92.3	85.1	71.9	60.7	50.0
		MCWB	41.4	49.4	53.1	59.2	65.7	71.0	71.1	69.3	65.2	59.4	54.6	45.2
	10%	DB	41.0	51.0	60.4	68.7	77.9	85.0	88.2	89.6	81.5	67.7	55.7	45.6
		MCWB	38.7	46.4	50.9	57.0	64.2	69.6	70.4	68.8	65.0	57.5	52.4	42.5
	0.4%	WB	48.4	58.5	59.7	65.6	71.9	76.3	77.0	76.3	71.9	64.7	60.2	52.7
		MCDB	53.7	67.8	73.7	79.6	86.6	90.4	92.0	90.8	86.1	75.0	68.9	58.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.6	53.4	56.4	63.1	69.1	74.1	74.6	74.0	69.5	62.6	57.6	48.5
		MCDB	48.9	58.8	67.6	77.0	81.1	87.3	88.7	87.8	80.5	73.0	65.1	53.9
	5%	WB	41.6	49.9	53.6	60.6	67.2	72.3	73.2	72.4	67.4	60.7	55.6	45.9
		MCDB	44.1	55.9	63.5	72.4	78.1	84.8	86.0	85.5	78.7	71.1	60.2	49.9
	10%	WB	39.0	46.8	51.2	58.3	65.2	70.9	71.9	70.9	66.0	58.7	52.9	43.1
		MCDB	40.7	50.7	58.7	66.8	75.2	82.5	84.0	83.6	77.3	66.2	56.3	45.6
Mean Daily Temperature Range	5% DB	MDBR	12.5	16.7	19.2	21.7	21.2	22.1	22.9	25.2	23.0	20.9	13.5	11.8
		MCDBR	19.1	26.1	29.7	31.4	27.9	26.9	28.1	30.3	29.8	29.8	23.9	21.2
		MCWBR	14.1	17.1	17.0	16.0	12.2	9.9	9.1	9.3	10.5	15.3	13.9	14.9
	5% WB	MCDBR	17.2	24.3	28.8	29.5	24.9	24.6	23.6	25.0	25.1	26.6	20.4	20.8
		MCWBR	13.0	16.3	16.3	15.0	11.3	9.9	8.6	8.8	9.6	14.0	11.7	14.6
Clear-Sky Solar Irradiance	taub		0.348	0.379	0.418	0.470	0.445	0.453	0.463	0.476	0.441	0.405	0.373	0.352
	taud		2.280	2.210	2.152	2.023	2.171	2.187	2.165	2.124	2.186	2.271	2.313	2.292
	Ebn at Noon		239	252	257	252	264	261	257	248	247	240	228	225
	Edh at Noon		28	35	42	52	46	46	46	47	41	33	27	25
All-Sky Solar Radiation	RadAvg		468	725	1087	1506	1863	2042	2067	1849	1345	872	497	380
	RadStd		70	94	104	154	150	139	137	112	104	98	70	54

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	+0.83	N/A	N/A	N/A	+0.74	N/A	N/A	-256	+124	N/A	

Nomenclature: See separate page